# Project 1: Collaborative Filtering

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Collaborative Filtering ⟺ Filling in missing values of a matrix

The matrix stores ratings, so we can recommend items to users

$$\begin{bmatrix} 3 & ? & ? & 5 & 1 & ? \\ ? & 1 & ? & 4 & 2 & 1 \\ 5 & 1 & ? & ? & ? & ? \\ 4 & 2 & 2 & ? & 2 & 4 \end{bmatrix}$$

# A paper recommendation system

**Two** sources of data:

- Sparse matrix of scientist-paper ratings
- Wishlists of papers they want to read

Task: Predict unobserved ratings

Metric: Root mean squared error

# Some dataset facts

- Rating scale is 1-5 stars
- 10,000 scientists and 1,000 papers
- Approximately 11% of ratings are observed
- On average, scientists have ~34 papers on their wishlist

# Getting started on the student cluster

1. Download the notebook `collaborative_filtering_intro.ipynb`
2. `scp collaborative_filtering_intro.ipynb <nethz>@student-cluster.inf.ethz.ch:~`
3. Start the Jupyter notebook server
    a. open https://student-jupyter.inf.ethz.ch/
    b. Set “Environment path” to `/cluster/courses/cil/envs/collaborative_filtering/bin`
    c. Set “Modules to add” to `cuda/12.6`
    d. Press “Start”
    e. Wait a few minutes for the server to start up

*Important:*
*Make sure to always stop your server when you are done because you only get a limited amount of GPU time.*

# Starting ideas: (Iterative) SVD

- You have seen SVD in lectures
- Assumes low-rank matrix

- Iterative SVD
  a. Impute matrix (e.g. mean)
  b. Compute SVD
  c. Reconstruct matrix from SVD
  d. Use reconstruction as imputation
  e. repeat

# Starting ideas: SVD++

- excellent and well-performing baseline for many collaborative filtering tasks
- Useful for ensembles
- Read about it
  - Factorization Meets the Neighborhood: a Multifaceted Collaborative Filtering Model, Yehuda Koren, KDD’08.

# Alternating Least Squares

- Discussed in the lecture
- Adding a Frobenius norm regularization on the training loss
- Can be minimized via alternatively minimizing each decomposed matrix

# Neural Methods

- Neural Collaborative Filtering: https://arxiv.org/pdf/1708.05031.pdf
- Deep Matrix Factorization: https://www.ijcai.org/proceedings/2017/0447.pdf

- Check the literature and consider more!

# Additional considerations

- Tuning hyperparameters based on cross validation
- Ensembles almost always work well
- Don’t forget to use the wishlist information. **Be creative!**